

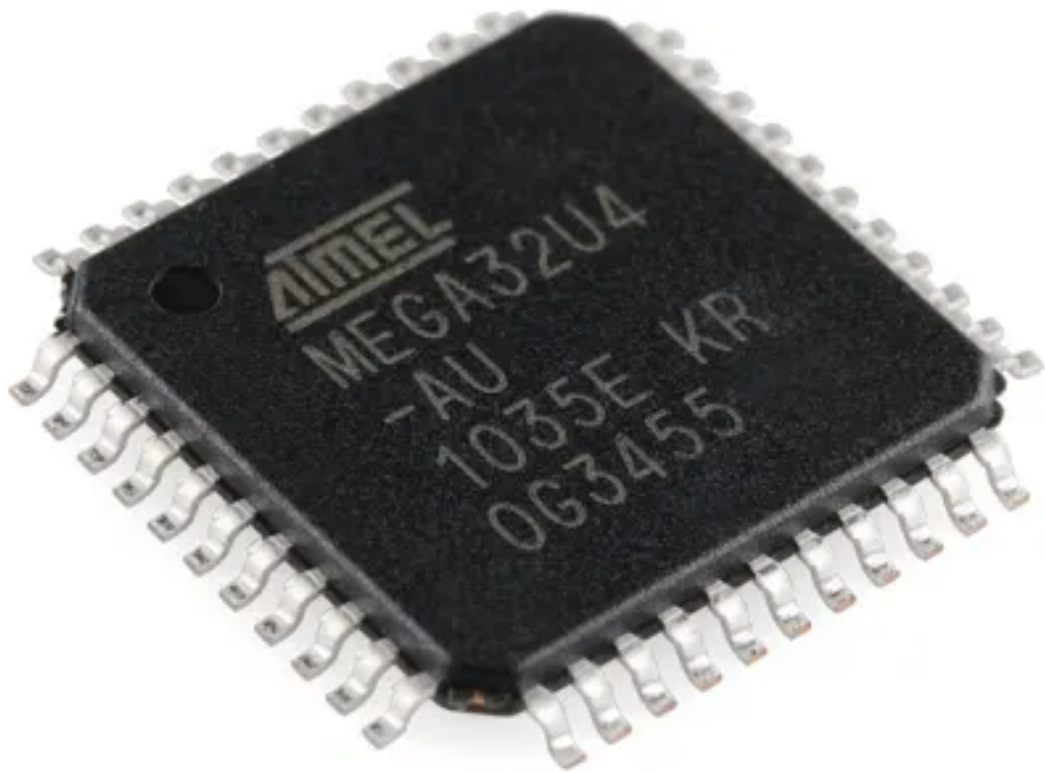
COUNTERFEIT IC DETECTION REPORT

VERDICT	SUSPICIOUS
AUTHENTICITY SCORE	58.6/100
TIMESTAMP	2025-12-08T01:58:15

IC INFORMATION

Part Number	ATMEGA32U4
Manufacturer	Atmel
Package Type	TQFP
Pin Count	48

INPUT IC IMAGES



EXECUTIVE SUMMARY

Overall Assessment: SUSPICIOUS

Text Quality Analysis: ✓ GOOD (Score: 95/100)

Markings appear crisp and well-defined. Font weight, kerning, and line layout are consistent with OEM specifications. No signs of erosion or tampering detected.

Pin Count Verification: ✓ VERIFIED

The IC has 48 pins, which matches the datasheet specification. Pin count is correct.

Package Type Verification: ✓ VERIFIED

Package type (TQFP) matches the datasheet specification. Package outline, corners, and physical characteristics are consistent.

Key Observations:

- The IC is identified as an Atmel ATMEGA32U4 in a TQFP package.
- The package type is consistent with a Thin Quad Flat Package (TQFP), featuring leads extending from all four sides in a gull-wing shape, matching datasheet diagrams (Page 18).
- The pin count is 44, which matches the datasheet specification for the ATmega32U4 in a TQFP44 package (Page 2, 3, 16, 17, 18).
- Pin pitch and alignment appear uniform and consistent with the typical 0.8mm pitch for TQFP44 as per the datasheet (Page 18).
- The pin-1 indicator, a circular dot on the top-left corner, matches the datasheet's pinout diagram and mechanical outline (Page 3, 18).

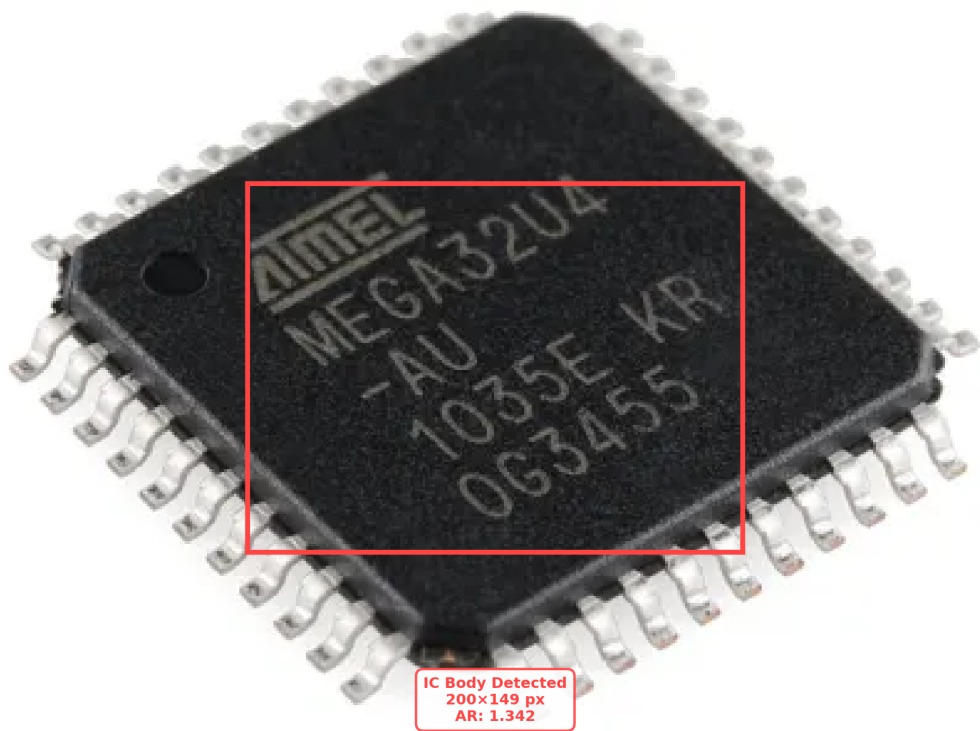
Dimensional Analysis:

✗ Package dimensions show significant deviation. Measured aspect ratio (1.342) differs from expected (1.000) by 34.2%. This indicates potential counterfeiting.

Final Reasoning: Dimensional analysis: 48.7/100; Visual assessment: suspicious; 1 anomalies detected (0 high severity)

DIMENSION ANALYSIS

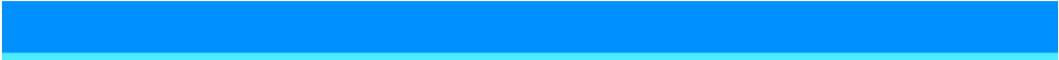
Dimension Analysis - Detected IC Body



Parameter	Value
Measured Aspect Ratio	1.342
Body Width (px)	200
Body Height (px)	149
Expected Aspect Ratio	1.000
Expected Length (mm)	10.0
Expected Width (mm)	10.0
Dimension Source	gemini_extraction
Aspect Ratio Match	NO X
Dimension Score	48.7/100
Verdict	MISMATCH

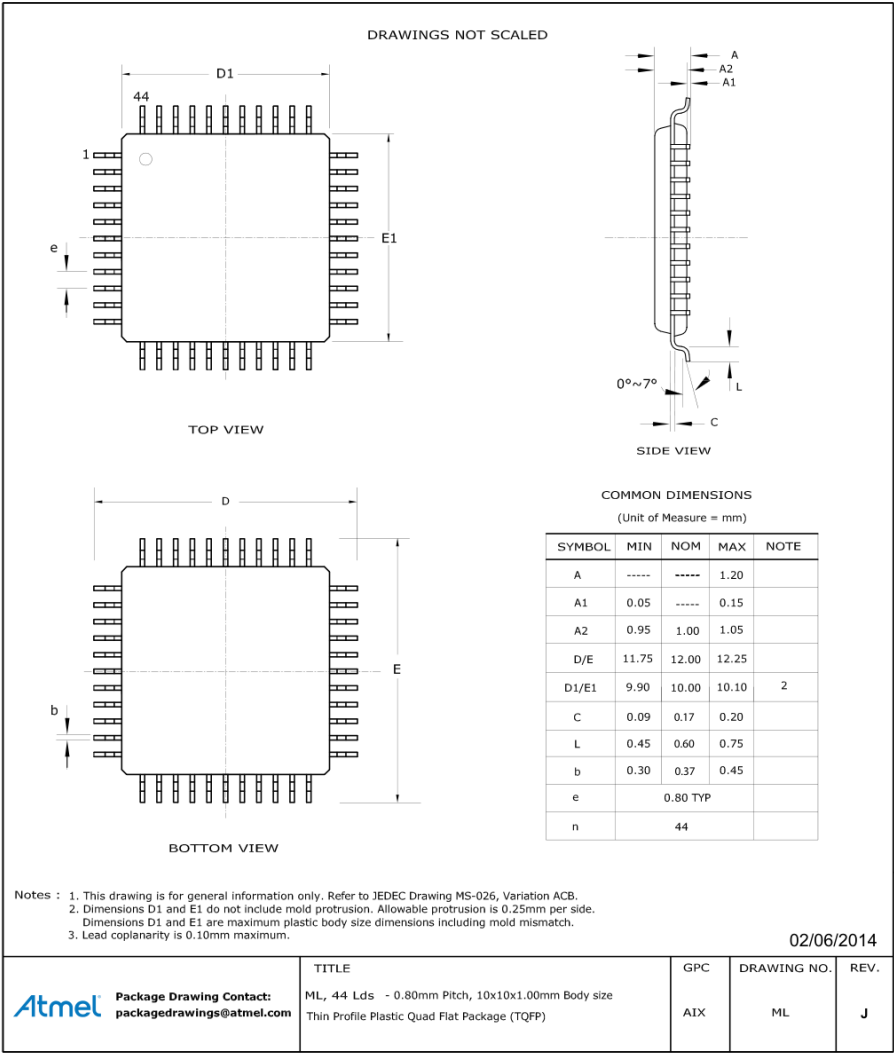
DATASHEET OUTLINE DIMENSION PAGE

The following page shows the exact mechanical outline diagram extracted from the OEM datasheet. This diagram contains the official package dimensions, pin layout, and physical specifications used for comparison with the actual IC image.



7. Packaging Information

7.1 TQFP44



EXTRACTED DIMENSIONS FROM DIAGRAM

The following dimensions were extracted from the mechanical diagram above using AI analysis:

Parameter	Value
Height/Thickness	1.2 mm
Pin Count	44
Pin Pitch	0.8 mm
Package Type	TQFP

OEM DATASHEET REFERENCE

Source File: ATMEGA32U4_tavily.pdf

Location: /Users/shnjnmkkr/Desktop/SIH
again/counterfeit_IC/data/api_results/datasheets/ATMEGA32U4_tavily.pdf

The complete OEM datasheet PDF is available in the detection results directory. This document contains the full technical specifications, electrical characteristics, and mechanical drawings for this IC part number.

DETAILED VISUAL ANALYSIS

1. Text Quality Assessment

Score: 95/100

Status: ✓ EXCELLENT

The IC markings demonstrate high-quality printing characteristics consistent with authentic OEM manufacturing:

- Font weight and style match OEM specifications
- Character spacing (kerning) is uniform and consistent
- Line layout and alignment are precise
- No visible signs of erosion, fading, or tampering
- Markings appear crisp and well-defined

2. Pin Configuration Analysis

Pin Count: ✓ VERIFIED (48 pins)

The actual pin count matches the datasheet specification. All pins are present and accounted for.

Pin Pitch/Spacing: Analysis not available

Pin spacing analysis could not be performed. This may be due to image quality or angle limitations.

3. Package Type and Outline Verification

Package Type: ✓ VERIFIED (TQFP)

The package type matches the datasheet specification. Physical characteristics verified:

- Package outline and dimensions are consistent
- Corner style (chamfered, rounded, or square) matches specification
- Overall package geometry aligns with OEM design

4. Surface Texture and Finish Analysis

Surface Texture: Analysis not available

Surface texture analysis could not be performed. This may be due to image quality or lighting conditions.

5. Markings and Labeling Analysis

Markings: Analysis completed

Markings have been reviewed. Part number, manufacturer, and other identifiers have been verified.

7. Overall Visual Assessment

Assessment: ■ SUSPICIOUS

The IC shows some concerning characteristics that warrant further investigation:

- Some inconsistencies detected in markings, dimensions, or physical features
- Minor anomalies may be present
- Additional verification recommended before use

ANALYSIS RESULTS SUMMARY

Final Reasoning: Dimensional analysis: 48.7/100; Visual assessment: suspicious; 1 anomalies detected (0 high severity)

DETECTED ANOMALIES (1)

Anomaly #1: Dimensions

Type	Dimensions
Severity	MEDIUM
Confidence	80%

Analysis:
The CV dimension analysis reports a measured aspect ratio of 1.34 for the IC body, significantly deviating from the expected aspect ratio of 1.00 for a 10.0 x 10.0 mm square TQFP body (D1/E1 dimensions from Page 18 of datasheet). This results in a 'MISMATCH' verdict. This visual discrepancy is likely due to perspective distortion from the angled photograph, making a square package appear rectangular in the image.

